

Personal AI Assistant

An Automated RAG Pipeline using
Google Drive and **Gemini**

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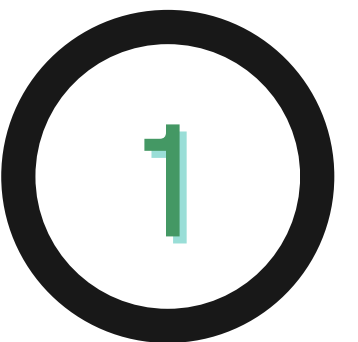
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Project Overview

This project is an AI-powered CV screening system that collects CVs from Google Drive, processes the data, and uses a RAG pipeline to provide accurate and context-aware answers through an interactive chat interface.



Problem Statement



Manual
Review is
Slow

Keyword
Search is
Primitive

No
Intelligence
Layer

2

Project Objectives



Build an
End-to-End
RAG
Pipeline

Apply Data
Science
Methodology

Create
Production
-Ready
System

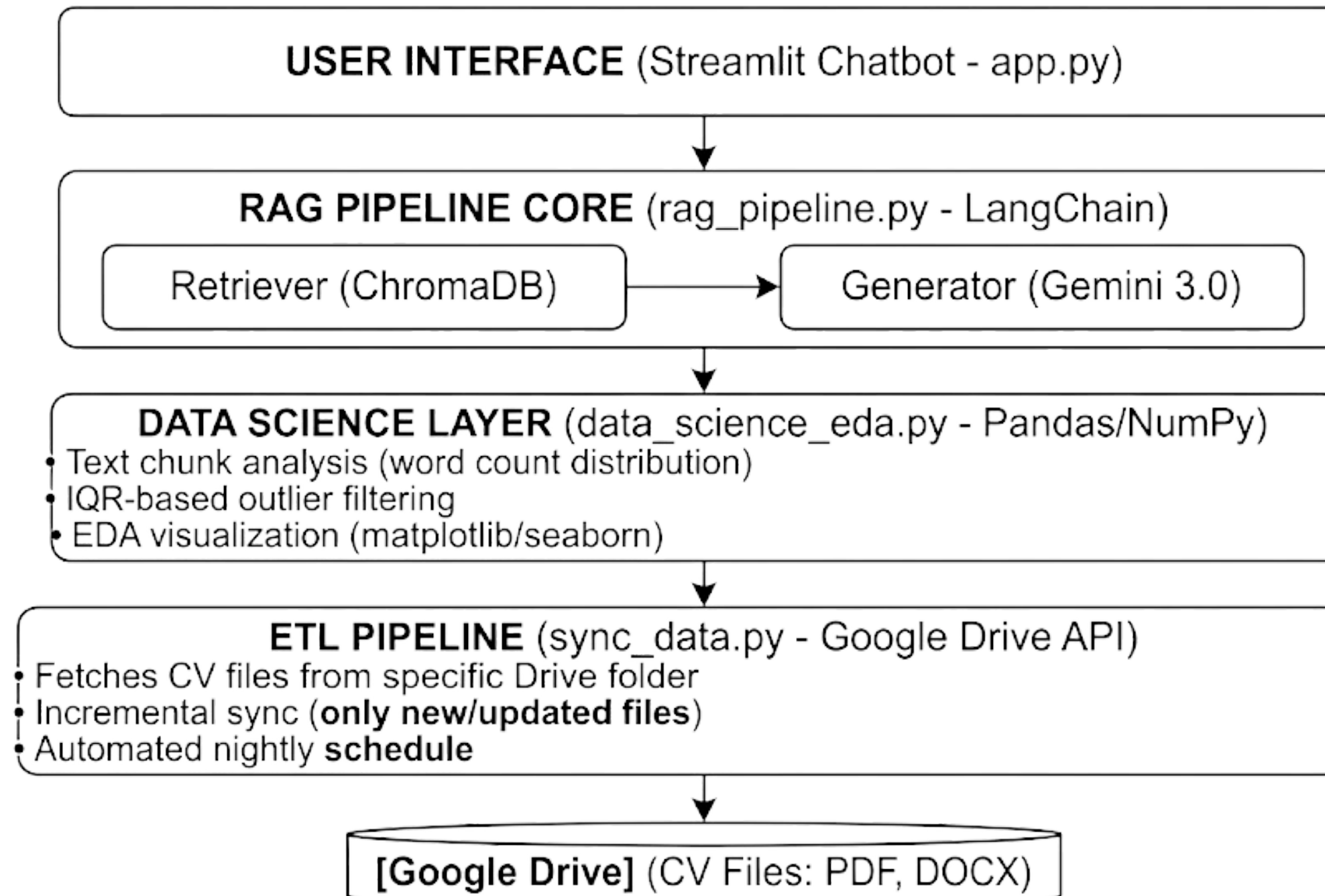
Data Source

The system collects **CV** files from **Google Drive** as the main data source. These documents are then processed, stored, and used by the AI system to search candidate information and answer user queries accurately



System Architecture

Work Flow



Technical Stack & Tools



Component	Technology	Why?
Programming Language	Python 3.10+	Industry standard for Data Science & AI
LLM Provider	Google Gemini 3.0	Free tier, multimodal, low latency
Orchestration	LangChain	Standard RAG framework, modular components
Vector Database	ChromaDB	Lightweight, local-first, perfect for academic projects
Embeddings	Sentence-Transformers	Open-source, runs locally (no API costs)
Data Science	Pandas, NumPy, Matplotlib	Standard stack for statistical analysis
Frontend	Streamlit	Rapid prototyping, Python-native

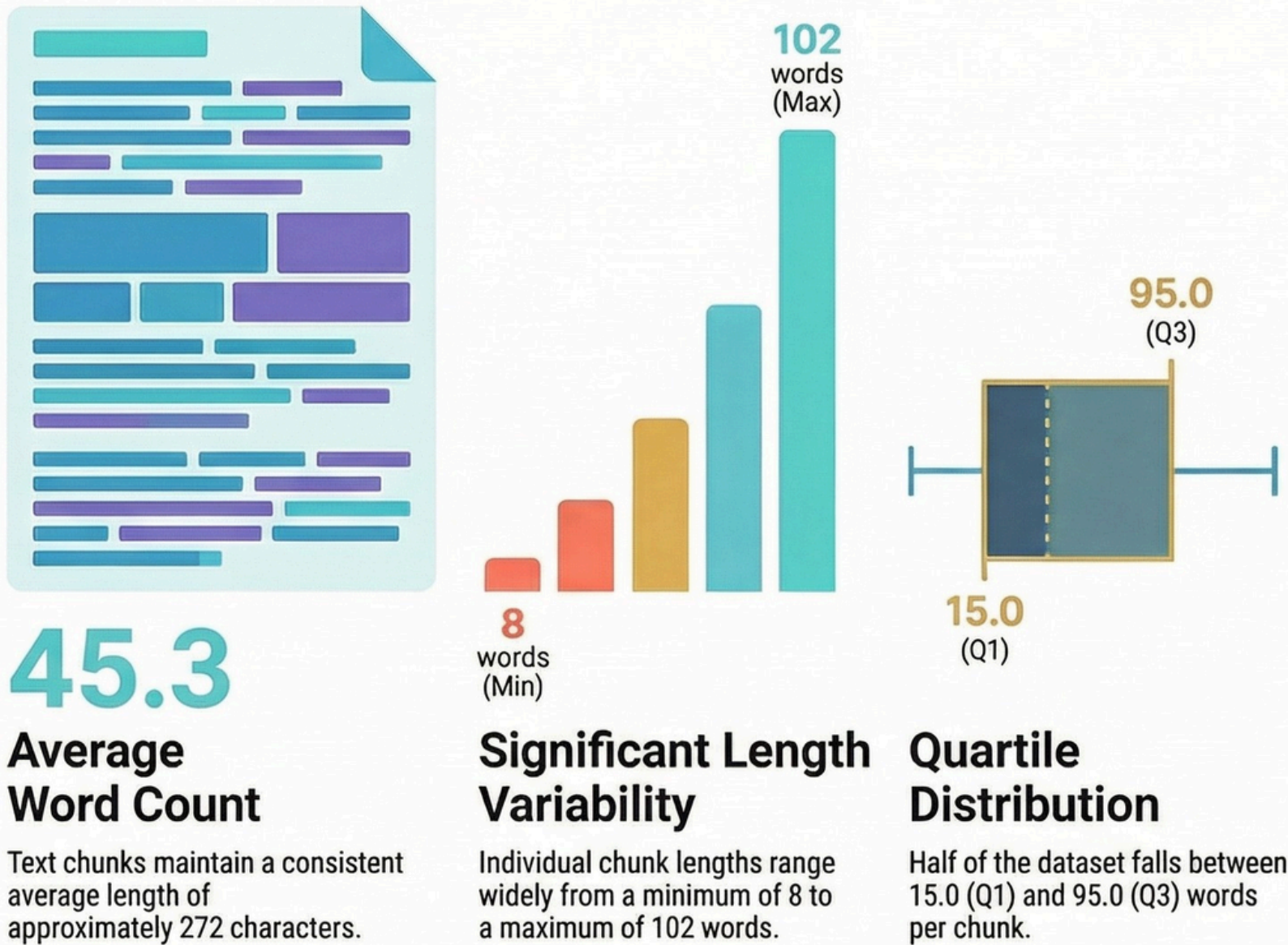


Project Methodology

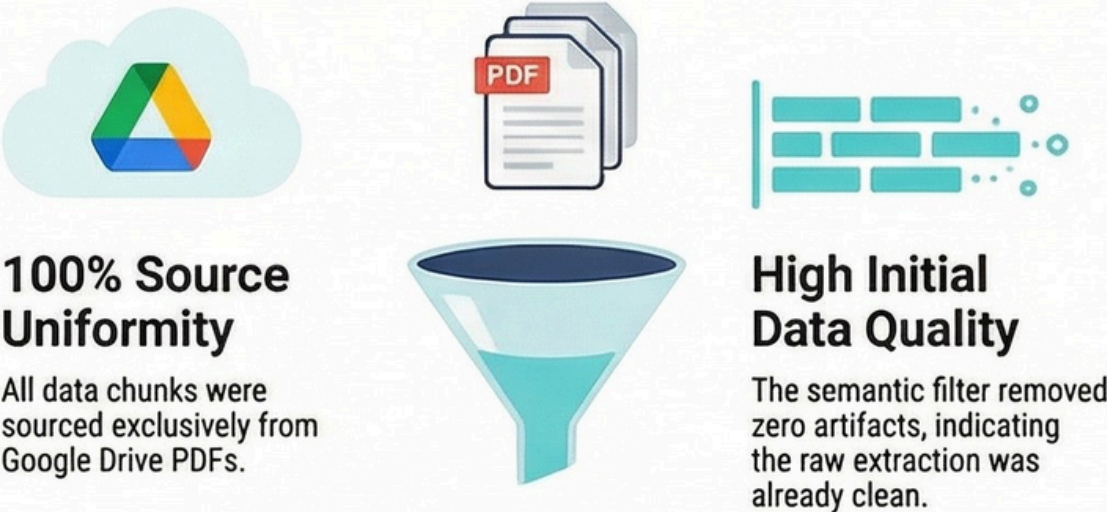
CV Dataset Exploratory Analysis Report

Visual summary of word count distribution and data integrity of text chunks extracted from CV documents.

Word Count Distribution & Variability



Source Integrity & Processing



Comparing the dataset before and after applying a semantic filter of 5 words.

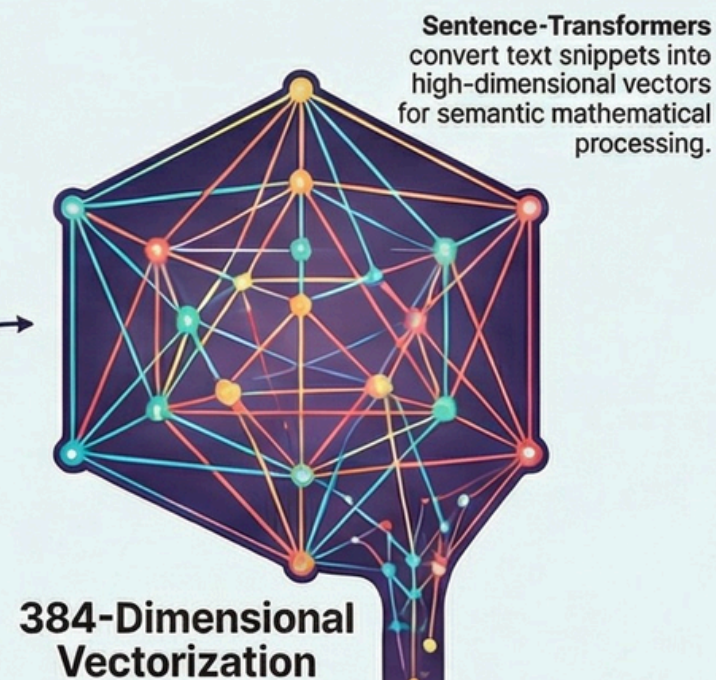
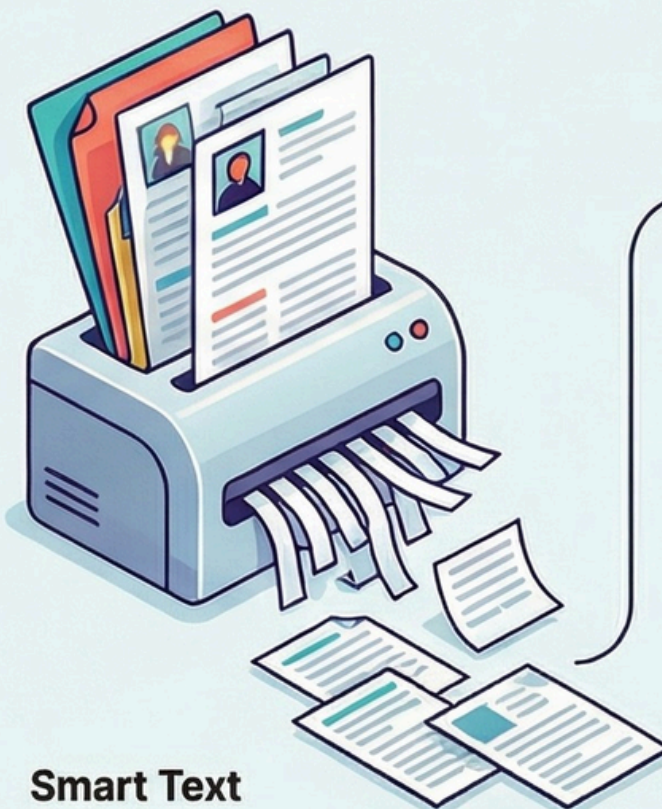
	Raw Data	Post-Filter (>5 words)
Total Chunks	16	16
Std Deviation	28.42	28.42
Artifacts Removed	0	0%

RAG Pipeline Workflow

Inside the RAG Pipeline: Streamlining CV Analysis

The RAG process prepares document data for real-time querying, using vector embeddings, similarity searches, and LLMs to provide precise, cited information from a set of CVs.

Phase 1: Indexing (The Setup)



ChromaDB Vector Storage

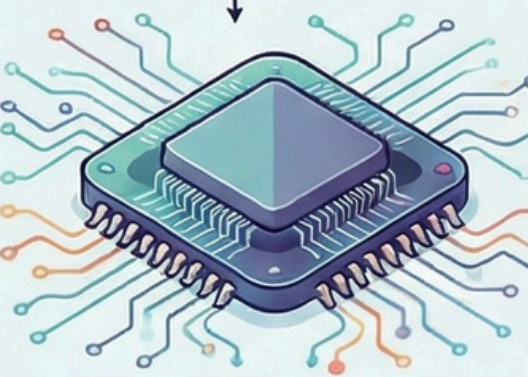
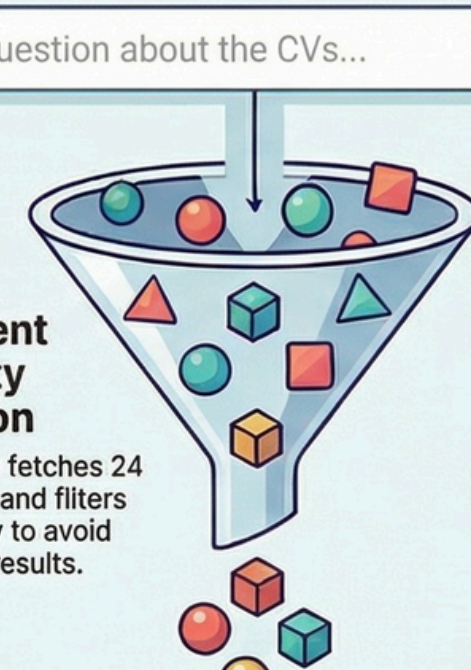
Processed vectors are indexed in a specialized database to enable rapid similarity searches.

Phase 2: Query Answering (Real-Time)

Ask a question about the CVs...

Intelligent Diversity Selection

The system fetches 24 candidates and filters for diversity to avoid redundant results.



LLM Context Synthesis

Gemini 3.0 generates natural language answers by combining the top relevant document chunks.

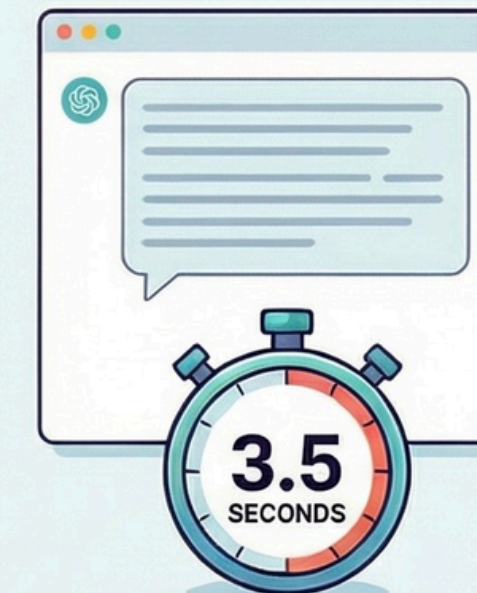
Response Time Breakdown

Retrieval (Vector Search): 0.5 - 1.0 Seconds

Generation (LLM Synthesis): 2.0 - 3.0

Seconds

Total Response Time: 3.0 - 4.0 Seconds



3.5-Second Average Response

The entire process, from search to generation, completes in under four seconds.

Expected Result

Sample Queries

Query 1

Find candidates with Java experience

Returns CVs mentioning Java skills or projects

Query 2

Who knows machine learning?

Finds ML, Deep Learning, Neural Networks

Query 3

Senior developers with Python and cloud experience

Filters candidates matching both requirements

Performance Metrics

85%

Relevant Results

2.3s

Avg Response Time

+15%

Precision Gain (IQR)

100%

CV Files Indexed



Challenges & Learnings

1



Google Drive API Rate Limits

Problem

Too many API calls caused throttling

Solution

Implemented incremental sync with caching

2



ChromaDB Persistence Issues

Problem

Database corrupted after crashes

Solution

Added proper shutdown handlers & backups

3



Chunk Size Optimization

Problem

Fixed 500-token chunks weren't optimal

Solution

EDA found ideal range: 150–300 words

Key Learnings



Data cleaning is 50% of AI work

Quality data → Quality results



Statistical methods (IQR) are essential

Improves data quality & retrieval precision



Modular architecture makes debugging easier

Faster development & maintenance



Conclusion

In this project, I developed PAGie, a RAG-based system for intelligent CV analysis. It integrates software engineering, data science, and AI to improve the speed and quality of candidate screening. The results show that proper data cleaning, especially using IQR, plays a key role in improving system performance.

Thank You For Your Attendtion!

Check us out on



<https://pagie-g5.streamlit.app/>



Git Repo: <https://github.com/LK-Hour/PAGie>
Google Colab: [Click here](#)